EXTENSION BY ZERO REFLECTS EXACTNESS OF COMPLEXES OF MODULES ON AN OPEN OF A RINGED SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites two related extension-by-zero lemmas, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Exactness of extension by zero and its construction. Stacks Project, Modules on Sites, Lemma lemma-extension-by-zero-exact (tag 09B4) and Lemma lemma-extension-by-zero (tag 09B5). Provides exactness of j_{U!} and the sectionwise description of the adjunction unit.

Source reference. Original lemma 03DJ: sources/stacks-chow/tags.txt, source lines 4422–4422. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03DI: sources/stacks-chow/tags.txt, source lines 4421–4421. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $U \in \mathrm{Ob}(\mathcal{C})$. A complex of $\mathcal{O}_U$-modules $\mathcal{G}_1 \to \mathcal{G}_2 \to \mathcal{G}_3$ is exact if and only if $j_{U!}\mathcal{G}_1 \to j_{U!}\mathcal{G}_2 \to j_{U!}\mathcal{G}_3$ is exact as a sequence of $\mathcal{O}$-modules.*

Proof. We already know that $j_{U!}$ is exact, see Lemma 03DJ. Thus it suffices to show that $j_{U!} : \mathit{Mod}(\mathcal{O}_U) \to \mathit{Mod}(\mathcal{O})$ reflects injections and surjections.

For every $\mathcal{G}$ in $\mathit{Mod}(\mathcal{O}_U)$ we have the unit $\mathcal{G} \to j_U^* j_{U!}\mathcal{G}$ of the adjunction. We claim this map is an injection of sheaves. Namely, looking at the construction of Lemma 03DI we see that this map is the sheafification of the rule sending the object V/U of $\mathcal{C}/U$ to the injective map

$$\mathcal{G}(V/U) \longrightarrow \bigoplus\nolimits_{\varphi \in \mathrm{Mor}_{\mathcal{C}}(V,U)} \mathcal{G}(V \xrightarrow{\varphi} U)$$

given by the inclusion of the summand corresponding to the structure morphism $V \to U$. Since sheafification is exact the claim follows. Some details omitted.

If $\mathcal{G} \to \mathcal{G}'$ is a map of $\mathcal{O}_U$-modules with $j_{U!}\mathcal{G} \to j_{U!}\mathcal{G}'$ injective, then $j_U^* j_{U!}\mathcal{G} \to j_U^* j_{U!}\mathcal{G}'$ is injective (restriction is exact), hence $\mathcal{G} \to j_U^* j_{U!}\mathcal{G}'$ is injective, hence $\mathcal{G} \to \mathcal{G}'$ is injective. We conclude that $j_{U!}$ reflects injections.

Let $a : \mathcal{G} \to \mathcal{G}'$ be a map of $\mathcal{O}_U$-modules such that $j_{U!}\mathcal{G} \to j_{U!}\mathcal{G}'$ is surjective. Let $\mathcal{H}$ be the cokernel of a. Then $j_{U!}\mathcal{H} = 0$ as $j_{U!}$ is exact. By the above the map $\mathcal{H} \to j_U^* j_{U!}\mathcal{H}$ is injective. Hence $\mathcal{H} = 0$ as desired. □

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